MCQ A · circle and functions

After S76; k denotes an integer (positive or negative).

For each question, only one option is correct. Just consult the solutions.

1. $\cos(-x)$ is:

A. $-\cos x$ B. $\cos x$

C. $\sin x$ D. 0

2. The smallest strictly positive period of $x \mapsto \sin(3x)$ is:

A. 2π B. 3π

C. $\frac{2\pi}{3}$ D. $\frac{\pi}{3}$

3. The solutions of $\sin x = 0$ are:

A. $x = \frac{\pi}{2} + 2k\pi$ B. $x = k\pi$

C. $x = 2k\pi$ only D. $x = \pi/4$

4. The derivative of $\cos x$ is:

A. $\sin x$ B. $-\sin x$

C. $-\cos x$ D. $\tan x$

5. On $[0;\pi]$, the sine is:

A. always negative B. positive or zero

C. Always increasing D. always zero

MCQ B · equations and function study

After S79.

For each question, only one option is correct. Justify your choice orally before consult the solutions.

6. Solve $\cos x = 1$ on $[0;2\pi]$ gives:

A. 0 and 2π B. π

C. $\frac{\pi}{2}$ D. a single solution

7. To solve $\sin x \geq \frac{1}{2}$, in particular we need:

A. choose the correct arc, then intersect with the interval required B. divide by $\sin x$

C. use a binomial D. forget the endpoints

8. A function of the form $a + b\cos(\omega t)$, with $\omega \neq 0$ has amplitude:

A. a B. b

C. $|b|$ D. ω

9. If $f'(x) = \cos x$, f is decreasing on:

A. $]0;\frac{\pi}{2}[$ B. $]\frac{\pi}{2};\frac{3\pi}{2}[$

C. $]\frac{3\pi}{2};2\pi[$ D. $[-\frac{\pi}{2};\frac{\pi}{2}]$

10. Two solutions differing by 2π represent:

A. distinct angles but the same point on the circle B. A contradiction

C. opposite sines D. forbidden values

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\sin^2 x + \cos^2 x = 1$ for every real x.
2. $\sin x = \frac{1}{2}$ has only one solution on $[0; 2\pi]$.
3. $\sin(a+b) = \sin a + \sin b$ for all real numbers a, b.

Review of prior work

I can read the circle, exploit parity and period, solve with an interval and study a function. **À**

to watch out for: forgetting an endpoint solution; confusing a reading from a graph with a proof of an identity.